

Package ‘MultFourier’

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Type Package

Title Computes the p-values for Multinomial Goodness-of-fit tests

Version 1.0

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Description Computes the p-value for an exact Multinomial, Pearson's Chi-Squared, log-likelihood ratio, or Power-Divergence tests. The main function, `pval_fourier`, uses a Fourier series expansion. Also, the library provides an exhaustive method `pval_exhaustive`, which evaluates all elements within the support of the Multinomial distribution. The function `pval_flexible` uses either of these methods, depending on the size of the instance.

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Description

Computes the p-value for an exact Multinomial, Pearson's Chi-Squared, log-likelihood ratio, or Power-Divergence tests. The main function, `pval_fourier`, uses a Fourier series expansion. Also, the library provides an exhaustive method `pval_exhaustive`, which evaluates all elements within the support of the Multinomial distribution. The function `pval_flexible` uses either of these methods, depending on the size of the instance.

Details

The `MultFourier` package provides efficient methods for computing p-values in multinomial tests, including both exact and approximate approaches. The exact p-value is computed using an exhaustive algorithm, which is recommended for small datasets. For larger datasets, an approximation is provided using a Fourier series algorithm, which is computationally efficient and provides control over precision through parameters such as `max_terms`, `rel_eps`, and `undersampling`.

The package includes the following methods:

- `pval_exhaustive`: Computes the exact p-value by enumerating all possible outcomes. Suitable for small datasets.
- `pval_fourier`: Approximates the p-value using a Fourier series expansion. Efficient for large datasets.
- `pval_flexible`: Automatically selects between the exhaustive and Fourier series methods based on the dataset size.

Key features:

- Control over precision through parameters such as `max_terms`, `rel_eps`, and `undersampling`.
- Support for multiple test statistics, including logP, Pearson's chi-square, and log-likelihood ratio.
- Verbose output for debugging and analysis.

Author(s)

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References

- Thraves, C. and Subiabre, F. et al. (Year): "Title of the Paper". Journal Name, Volume(Issue), Pages.

See Also

- Report bugs at <https://github.com/PabRvss/MultFourier/issues>

print.multfourier	<i>Print method for multfourier objects</i>
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Description

Print method for multfourier objects

Usage

```
## S3 method for class 'multfourier'
print(x, ...)
```

Arguments

x	Object of class "multfourier" returned by pval_fourier, pval_exhaustive, or pval_flexible.
...	Additional arguments (currently unused).

Value

Invisibly returns the input object.

pval_exhaustive	<i>Exact multinomial p-value computation via iterative bisection</i>
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Description

Exact multinomial p-value computation via iterative bisection

Usage

```
pval_exhaustive(
  x,
  p = rep(1/length(x), length(x)),
  stat = c("llr", "pd", "chi2", "pmf"),
  lambda = 1,
  n_threads = 1,
  max_time = NULL,
  engine = c("speedup", "standard"),
  verbose = FALSE,
  ...
)
```

Arguments

x	Numeric vector of observed counts for each category.
p	Numeric vector of theoretical probabilities under the null hypothesis.
stat	Test statistic: "llr", "pd", "chi2", or "pmf". Default is "llr".
lambda	Real parameter for Power Divergence when stat = "pd". Default is 1. Ignored if stat != "pd".
n_threads	Number of execution threads (integer >= 1). Default is 1.
max_time	Maximum execution time in seconds (optional).
engine	Computational engine: "speedup" (default, uses pruning bounds via fast_exhaustive) or "standard" (iterative bisection in last categories). Default is "speedup".
verbose	Logical; if TRUE, prints progress information. Default is FALSE.
...	Additional internal parameters passed to the C engine.

Value

Returns an S3 object of class "multfourier" with the following attributes:

- x** The input vector of observed realizations for each category.
- p** The input vector of probabilities for each category under the null hypothesis.
- pval** The computed p-value.
- stat** A string describing the test statistic used.
- time** The total execution time of the algorithm in seconds.
- status** The final status ID of the algorithm upon completion (0: Converged, 1: Maximum time reached).
- message** The finishing status displayed as a human-readable message.
- method** A string with value "exhaustive".

pval_flexible

Multinomial p-value computation via flexible method

Description

Evaluates whether the dimension of the multinomial support allows exact enumeration (iterative bisection) or routes to the Fourier inversion series method.

Usage

```
pval_flexible(
  x,
  p = rep(1/length(x), length(x)),
  stat = c("llr", "pd", "chi2", "pmf"),
  lambda = 1,
  undersampling = 1,
  rel_eps = 0.001,
  B = 30,
  max_terms = 10000,
  max_time = NULL,
  avg_window = 0,
  avg_flat = FALSE,
  n_threads = min(2L, parallel::detectCores()),
  precision = c("double", "double-double"),
  precompute = TRUE,
  engine = c("speedup", "standard"),
  verbose = FALSE,
  ...
)
```

Arguments

x	Numeric vector of observed counts for each category.
p	Numeric vector of theoretical probabilities under the null hypothesis (must sum to 1).
stat	Test statistic: "llr" (Log-Likelihood Ratio), "pd" (Power Divergence), "chi2" (Pearson's Chi-squared), or "pmf" (Multinomial Point Probability Mass). Default is "llr".
lambda	Real parameter for Power Divergence when stat = "pd". Default is 1. Ignored if stat != "pd".
undersampling	Undersampling factor (> 0) for the period in the Fourier series. Default is 1.
rel_eps	Relative convergence tolerance (>= 0). Default is 0.001.
B	Integer >= 0; number of consecutive terms within tolerance required for convergence. Default is 30.
max_terms	Maximum number of harmonic terms to evaluate in the series (positive integer). Default is 10000.
max_time	Maximum execution time in seconds (optional).
avg_window	Fraction of final partial sums to average (between 0 and 1). Default is 0.
avg_flat	Logical; if TRUE, uses a flat moving average; if FALSE, applies linear weighting towards the end. Default is FALSE.
n_threads	Number of execution threads (positive integer >= 1). Default is min(2L, parallel::detectCores()).
precision	Arithmetic precision: "double" (standard) or "double-double" (extended precision). Default is "double".

precompute	Logical; if TRUE, uses precomputed transforms for acceleration. Default is TRUE.
engine	Computational engine: "speedup" (default, using Poisson series, Newton saddlepoint search, $O(K)$ bounds, and Shanks extrapolation) or "standard" (classic polynomial convolution and bisection algorithms). Default is "speedup".
verbose	Logical; if TRUE, prints progress information. Default is FALSE.
...	Additional internal parameters passed to the C engine (e.g. <code>enum_cutoff = 7.4</code> , the threshold for $\log_{10}$ support size under which exact bisection is selected, corresponding to approximately 1 second of execution).

Value

Returns an S3 object of class "multfourier" with the following attributes:

- x** The input vector of observed realizations for each category.
- p** The input vector of probabilities for each category under the null hypothesis.
- pval** The computed p-value.
- Time_gamma** The time spent optimizing gamma in seconds.
- W** The effective width of the distribution support interval.
- stat** A string describing the test statistic used (e.g. "Log-Likelihood Ratio", "Pearson's Chi-Square", "Probability Mass Function", or "power-divergence ($\lambda = \dots$)").
- Gamma** The optimal gamma obtained in the first part of the method.
- n_terms** The number of terms of the Fourier sum evaluated.
- time** The total execution time of the algorithm in seconds.
- status** The final status ID of the algorithm upon completion:
 - 0: Converged.
 - 1: Maximum time reached.
 - 2: Maximum number of terms reached.
 - 3: Could not solve the optimization of gamma.
 - 4: Numerical instability detected (magnitude explosion).
- message** The finishing status displayed as a human-readable message.
- method** A string with value "fourier" or "exhaustive".

pval_fourier

Multinomial p-value computation via Fourier series inversion

Description

Multinomial p-value computation via Fourier series inversion

Usage

```

pval_fourier(
  x,
  p = rep(1/length(x), length(x)),
  stat = c("llr", "pd", "chi2", "pmf"),
  lambda = 1,
  undersampling = 1,
  rel_eps = 0.001,
  B = 30,
  max_terms = 10000,
  max_time = NULL,
  avg_window = 0,
  avg_flat = FALSE,
  n_threads = min(2L, parallel::detectCores()),
  precision = c("double", "double-double"),
  precompute = TRUE,
  engine = c("speedup", "standard"),
  verbose = FALSE,
  ...
)

```

Arguments

<code>x</code>	Numeric vector of observed counts for each category.
<code>p</code>	Numeric vector of theoretical probabilities under the null hypothesis (must sum to 1).
<code>stat</code>	Test statistic: "llr" (Log-Likelihood Ratio), "pd" (Power Divergence), "chi2" (Pearson's Chi-squared), or "pmf" (Multinomial Point Probability Mass). Default is "llr".
<code>lambda</code>	Real parameter for Power Divergence when <code>stat = "pd"</code> . Default is 1. Ignored if <code>stat != "pd"</code> .
<code>undersampling</code>	Undersampling factor (> 0) for the period in the Fourier series. Default is 1.
<code>rel_eps</code>	Relative convergence tolerance (≥ 0). Default is 0.001.
<code>B</code>	Integer ≥ 0 ; number of consecutive terms within tolerance required for convergence. Default is 30.
<code>max_terms</code>	Maximum number of harmonic terms to evaluate in the series (positive integer). Default is 10000.
<code>max_time</code>	Maximum execution time in seconds (optional).
<code>avg_window</code>	Fraction of final partial sums to average (between 0 and 1). Default is 0.
<code>avg_flat</code>	Logical; if TRUE, uses a flat moving average; if FALSE, applies linear weighting towards the end. Default is FALSE.
<code>n_threads</code>	Number of execution threads (positive integer ≥ 1). Default is <code>min(2L, parallel::detectCores())</code> .
<code>precision</code>	Arithmetic precision: "double" (standard) or "double-double" (extended precision). Default is "double".

precompute	Logical; if TRUE, uses precomputed transforms for acceleration. Default is TRUE.
engine	Computational engine: "speedup" (default, using Poisson series, Newton saddlepoint search, $O(K)$ bounds, and Shanks extrapolation) or "standard" (classic polynomial convolution and bisection algorithms). Default is "speedup".
verbose	Logical; if TRUE, prints progress information. Default is FALSE.
...	Additional internal parameters passed to the C engine (e.g. <code>enum_cutoff = 7.4</code> , the threshold for $\log_{10}$ support size under which exact bisection is selected, corresponding to approximately 1 second of execution).

Value

Returns an S3 object of class "multfourier" with the following attributes:

- x** The input vector of observed realizations for each category.
- p** The input vector of probabilities for each category under the null hypothesis.
- pval** The computed p-value.
- Time_gamma** The time spent optimizing gamma in seconds.
- W** The effective width of the distribution support interval.
- stat** A string describing the test statistic used (e.g. "Log-Likelihood Ratio", "Pearson's Chi-Square", "Probability Mass Function", or "power-divergence ($\lambda = \dots$)").
- Gamma** The optimal gamma obtained in the first part of the method.
- n_terms** The number of terms of the Fourier sum evaluated.
- time** The total execution time of the algorithm in seconds.
- status** The final status ID of the algorithm upon completion:
 - 0: Converged.
 - 1: Maximum time reached.
 - 2: Maximum number of terms reached.
 - 3: Could not solve the optimization of gamma.
 - 4: Numerical instability detected (magnitude explosion).
- message** The finishing status displayed as a human-readable message.
- method** A string with value "fourier".

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